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Ab Initio Calculations on Porphin

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Abstracts

Ab initio SCF calculations are reported for the porphin molecule. The positions of the central protons have been optimized, and the equilibrium geometry is found to be a linear NH · · · HN arrangement. The NH vibrational frequencies have been computed and are compared to experimentally measured quantities. Several low ionized states have also been studied in separate spin-restricted SCF calculations. The lowest state is found to have B_{1u} symmetry with an ionization potential of 8.0 eV.

Des résultats de calculs SCF *ab initio* pour la molécule de porphine sont rapportés. Les positions de protons centraux ont été optimisées, ce qui donne une géometrie d'équilibre linéaire NH · · · HN. Les fréquences vibratoires NH ont été calculées et comparées à des quantités expérimentales. On a étudié aussi plusieurs états bas ionisés dans des calculs SCF restreints par rapport au spin. L'état le plus bas est de symétrie B_{1u} avec un potentiel d'ionisation de 8.0 eV.

Ab initio SCF-Berechnungen für das Porphinmolekül werden beschrieben. Die Lagen der zentralen Protonen sind optimisiert worden, was eine lineare Gleichgewichtsgeometrie NH · · · HN gibt. Die Schwingungsfrequenzen für NH sind berechnet und mit Experimentalgrössen verglichen worden. Mehrere tiefe ionisierte Zustände sind auch mit spinbegrenzten SCF-Berechnungen studiert worden. Der tiefste Zustand hat B_{1u}-Symmetrie und ein Ionisierungspotential von 8.0 eV.

1. Introduction

Porphins and porphyrins have long attracted the interest of biochemists, mainly because of their biological activity. The free base porphin H_2P (cf. Figure 1) can be regarded as the parent molecule for such systems and has already been subject to several theoretical investigations [1–7]. So far these have all been at a semiempirical level. Such methods necessarily involve a certain amount of ambiguity, and it is hoped that *ab initio* calculations will allow more definite conclusions to be drawn.

The positions of the central protons in H_2P have often been disputed [8]. It would now seem to be well-established that the protons are arranged diametrically across the centre of the molecule [9]; there remain, however, several alternative geometrical possibilities within this restriction. The structure suggested in

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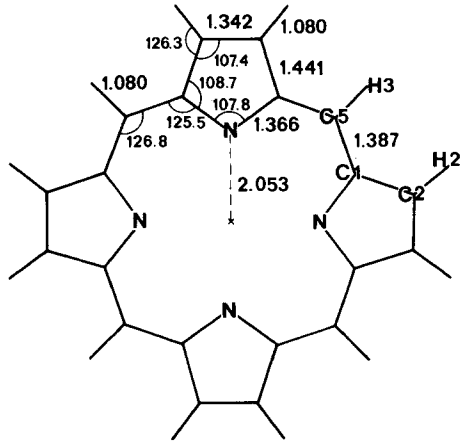

Figure 1. The molecular geometry used in the calculations. Bond distances in Å, angles in degrees.

Figure 2a involving a linear NH · · · HN arrangement will be referred to as Type A in the following. A situation where each proton is equally shared between two adjacent nitrogens as in Figure 2b would also be possible and is referred to as Type B. The intermediate situation between A and B, as shown in Figure 2c, with the protons covalently bonded to two opposite nitrogens and hydrogen-bonded to the remaining two is termed Type C. Finally, since a Type A structure would lead to a rather short H · · · H distance (~2.1 Å assuming normal N—H distances), distorted, nonplanar structures (Type D) should also be considered. One purpose of the calculations presented here has been to determine the relative stability of

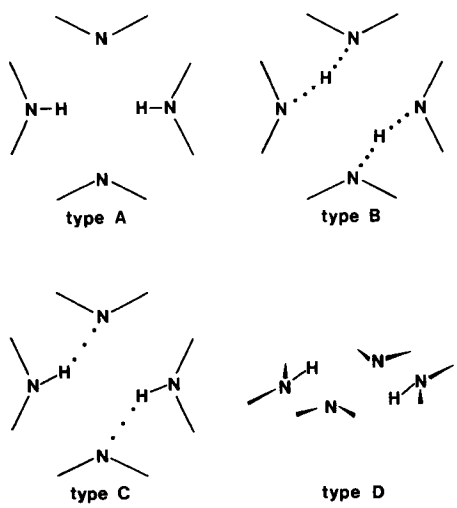

Figure 2. Illustration of different possible types of geometry for porphin.

these different possible structures, and especially the influence of hydrogen bonding.

It is an experimentally established fact that the effect of hydrogen bonding on IR frequences is generally much greater than its effect on the structure. The N—H vibrational frequencies have therefore been computed and compared to experimental values found in similar systems.

In addition to the calculations on the neutral molecule, the energies of the lowest ionized states have also been computed.

2. Computational Methods

The calculations have been performed with the *ab initio* MO-LCGO-SCF method, using the formalism described by Roothaan [10–12].

When performing LCAO calculations on large, symmetric molecules, it would be highly desirable to make use of the molecular symmetry in the integral evaluation as well as in the SCF section. In the present molecule, the Type A structures possess D_{2h} symmetry, and most of the atoms occur in fourfold positions. For these cases, only about 25% of the atomic integrals are unique in magnitude. The remaining 75% can be generated from these simply by multiplying by $+1$ or -1. Use of symmetry in the atomic integral computation can accordingly decrease the time and labour needed by a factor of four. Even greater gains can be achieved in the storage of the integrals; a problem which would seem to be a limiting factor for very large LCAO calculations. If the integrals are computed over a symmetry-adapted basis (and if the supermatrix formalism [10–12] is used), the number of $\mathscr{P}$- and $\mathscr{K}$-supermatrix elements to be stored will be reduced by a factor of nearly 30 in the present case, i.e., a decrease from $170 \cdot 10^6$ to $6 \cdot 10^6$ (if all $\mathscr{P}$- and $\mathscr{K}$-elements were stored, regardless of their magnitude). Besides simplifying the storage of data, this will speed-up the SCF calculation since the main part of the SCF time is spent in reading and processing the integral list. However, a straightforward computation of all atomic integrals, followed by a separate transformation step would not be feasible in large calculations since this would require the intermediate storage of all the atomic integrals and an inconvenient transformation procedure involving two large data sets. Instead, the calculation and transformation of integrals must be performed simultaneously. One method for doing this which has been found quite efficient for nondegenerate point groups (i.e., D_{2h} and any one of its subgroups) is described in the following.

A symmetry-adapted basis function $\phi_{\lambda a}$ transforming as the irreducible representation λ can be constructed from the atomic orbital χ_a as

$$\phi_{\lambda a} = N_{\lambda a} \sum_i C_{i\lambda}\, g_i \cdot \chi_a$$

Here, $\{g_i\}$ is a set of symmetry operators transforming χ_a to all symmetry-related sites in the molecule. For the group D_{2h} or its subgroups the coefficients $C_{i\lambda}$ are

$= +1$ or -1 with a proper choice of the normalization factor $N_{\lambda a}(C_{i\lambda} = -1$ if $\phi_{\lambda a}$ changes sign when operated on by g_i .) An electron-repulsion integral over symmetry-adapted functions can now be constructed:

$$I = \left\langle \phi_{\lambda a}(1)\phi_{\mu b}(1)\frac{1}{r_{12}}\phi_{\nu c}(2)\phi_{\pi d}(2)\right\rangle$$

$$= N\sum_{ijkl} C_{i\lambda}C_{j\mu}C_{k\nu}C_{l\pi}\left\langle g_i\cdot\chi_a(1)g_j\cdot\chi_b(1)\frac{1}{r_{12}}g_k\cdot\chi_c(2)g_l\cdot\chi_d(2)\right\rangle$$

Transforming the entire product to be integrated by any member of the group will of course not change the value of the integral. Making such a transformation with the operator g_i^{-1} will give the expression

$$I = N\sum_{ijkl} C_{i\lambda}C_{j\mu}C_{k\nu}C_{l\pi}\langle\chi_a, g_i^{-1}\cdot g_j\cdot\chi_b \mid g_i^{-1}\cdot g_k\cdot\chi_c, g_i^{-1}\cdot g_l\cdot\chi_d\rangle$$

Introducing $g_J = g_i^{-1}\cdot g_j$, $g_K = g_i^{-1}\cdot g_k$, $g_L = g_i^{-1}\cdot g_l$ the coefficients C will transform as $C_{j\mu} = C_{J\mu}\cdot C_{i\mu}$, $C_{k\nu} = C_{K\nu}C_{i\nu}$, $C_{l\pi} = C_{L\pi}C_{i\pi}$. The integral now becomes

$$I = N\sum_i C_{i\lambda}C_{i\mu}C_{i\nu}C_{i\pi}\sum_{JKL} C_{J\mu}C_{K\nu}C_{L\pi}\langle\chi_a, g_J\cdot\chi_b \mid g_K\cdot\chi_c, g_L\cdot\chi_d\rangle$$

If the product of the four irreducible representations λ, μ, ν and π is the totaly symmetric representation the product $C_{i\lambda}C_{i\mu}C_{i\nu}C_{i\nu}$ equals unity regardless of the index i. Other cases are trivial since the entire integral will then be zero by symmetry. Therefore, the entire summation over i is redundant and may be replaced by a constant multiplicative factor.

The algorithms outlined above provide considerable simplifications in calculations on large, symmetric molecules since recomputation of symmetry-redundant integrals is avoided and since symmetry-adapted integrals are obtained without an explicit transformation involving I/O. A computer program which makes use of the above ideas has been written and incorporated in the program system MOLECULE [13], and has been used in the present work. The main part of the calculations was carried out on the IBM 360/195 computer at the IBM Research Centre, San Jose. For the Type A geometries, the integral calculation required 28 min of CPU time for the porphin skeleton and a further 3 min for each geometry to add the necessary integrals involving the central hydrogens. The SCF calculations required 6 sec of CPU time per iteration for the ground state and 13 sec for the ions. The less symmetric cases, possessing only a centre of inversion, took 54 + 6 min for the integral computation, and 35 sec for each SCF iteration. The complete computation of 23 ground state geometries and 10 ionized states was done in 6 h 30 min.

3. Calculations and Results

In all calculations, the porphin moiety was assumed to have D_{4h} symmetry, and its geometry was not subject to changes when the positions of the central protons were varied. The geometry used for the porphin skeleton was taken from the X-ray diffraction investigation by Webb and Fleischer [14], after small adjustments to obtain perfect D_{4h} symmetry. Since hydrogen positions are poorly resolved in X-ray work, C—H distances of 1.08 Å were adopted. The atomic coordinates used are shown in Table I, and the bond distances and angles are displayed in Figure 1. It is of course necessary to keep the porphin skeleton rigid through the calculations in order to reduce the number of degrees of freedom to be varied. It is also of great value in the calculation of electron-repulsion integrals, which is by far the most time-consuming part of the calculations. All integrals over basis functions of the porphin skeleton are equal from case to case, and only the relatively few integrals involving the central hydrogens need to be recomputed for each geometry.

The basis functions used consisted of fixed linear combinations of Gaussians. For carbon and nitrogen, basis sets with 7 *s*- and 3 *p*-type functions were contracted to a minimal basis set. The orbital exponents and contraction coefficients used were those given by Roos and Siegbahn [14, 15]. For hydrogen, the three *s*-exponents given by Huzinaga [16] were contracted to one, except for the central hydrogens, where two contracted functions were used in order to allow for additional flexibility in the region of greatest interest. The hydrogen exponents were scaled by a factor 1.2, a procedure which has been found empirically to yield slightly better results than unscaled values. Thus, a total of 426 primitive Gaussians (GTO:s) were contracted to an effective basis set of 136 functions (CGTO:s). In all calculations, the two central hydrogen atoms have been related by an inversion centre. This reduces the number of degrees of freedom to be varied, and also the time and labour needed to compute each point in the potential hypersurface, as was discussed in the previous section. This was not considered a serious restriction in the optimization of the geometry since a completely unsymmetric equilibrium geometry would seem most improbable. The computed energies and corresponding

TABLE I. Atomic coordinates used in the calculations (in a.u.).

Atom	X	Y
N	3.880	0.0
C1	5.401	2.086
C2	7.999	1.268
C3	4.573	4.573
H2	9.645	2.474
H3	6.017	6.017

TABLE II. Calculated total energies and corresponding geometries. X, Y and Z are the coordinates of one of the symmetry-related protons and r is the N—H distance.

Nr	Type	X(a.u.)	Y(a.u.)	Z(a.u.)	r(Å)	E(a.u.)
1	A	2.5	0.0	0.0	.730	979.6019
2	A	2.1	0.0	0.0	.942	.8469
3	A	2.0	0.0	0.0	.995	.8618
4	A	1.9	0.0	0.0	1.048	.8630
5	A	1.8	0.0	0.0	1.101	.8522
6	A	1.7	0.0	0.0	1.154	.8318
7	A	1.6	0.0	0.0	1.207	.8040
8	A	1.5	0.0	0.0	1.259	.6672
9	A	1.0	0.0	0.0	1.524	.6055
10	C	2.0	0.2	0.0	1.000	.8597
11	C	2.5	0.5	0.0	.777	.6767
12	C	2.2	0.5	0.0	.928	.8214
13	C	2.0	0.5	0.0	1.029	.8497
14	C	1.5	0.5	0.0	1.289	.7637
15	C	2.0	1.0	0.0	1.127	.8059
16	C	1.5	1.0	0.0	1.366	.6672
17	B	1.0	1.0	0.0	1.613	.6732
18	D	2.0	0.2	0.2	1.006	.8584
19	D	2.0	0.5	0.2	1.035	.8483
20	D	2.0	0.0	0.5	1.025	.8530
21	D	2.0	0.2	0.5	1.035	.8503
22	D	2.0	0.5	0.5	1.063	.8386
23	D	2.2	0.5	0.5	.965	.8299

geometries are shown in Table II. In order to find the equilibrium geometry, a polynomial in r, y and z was fitted to the computed energy values. The energy minimum occurs for a Type A structure, i.e. $y = z = 0$, with an N—H distance of 1.025 Å. One might expect Type C distortions of the structure to be stabilized by hydrogen bonding, the equilibrium H $\cdots$ N distance being only 2.30 Å. This is not reflected in any structural effects, however. (The slight increase in NH distance as compared to free NH groups can hardly be regarded as significant considering the limited size of the basis set used.) Out-of-plane bending frequencies of >NH groups are quite sensitive to hydrogen bonding. The frequencies of free groups are around 500 cm^{-1}, but increase to 900–1000 cm^{-1} in the moderately strong hydrogen bond ($R_{NN} = 2.86$ Å) found in solid imidazole [17]. From the

computed energy values a similar value of 960 cm^{-1} for symmetric out-of-plane bending is derived. (The symmetry classification refers to the inversion centre.) The calculated symmetric stretching and symmetric in-plane bending frequencies are 3940 cm^{-1} and 1130 cm^{-1}, respectively, as compared to the typical values 3500 cm^{-1} and 1250 cm^{-1} for free >NH groups. The increase in the stretching frequency is not unexpected, since the situation for symmetric stretching is most unfavourable. (The asymmetric stretching frequency in porphin has been experimentally determined to be 3305 cm^{-1} [18].) The in-plane frequency compares well with the value 1190–1240 cm^{-1} typical for hydrogen-bonded cases. One can thus conclude that, from many aspects, the situation is similar to that in hydrogen-bonded cases, but the occurrence of two competing acceptors prevents any noticeable structural hydrogen bond effect. The term "bifurcated hydrogen bond" may be justified here.

For the geometry number 4, which is close to the equilibrium geometry, a series of spin-restricted SCF calculations was carried out for several ionized states. The results are presented in Table III. In Figure 3 the energy levels are compared

TABLE III. Orbital energies and ionization potentials for porphin (in eV).

Symmetry	ΔSCF	ε
B_{1u}	8.00	8.42
A_u	8.39	8.78
B_{3g}	10.41	11.02
B_{1u}	11.08	11.41
B_{2u}	11.24	12.76
A_g	11.82	12.76
B_{2g}	12.16	12.74
B_{2g}	12.35	12.58
B_{1u}	12.44	12.75
B_{3g}	12.62	12.89

to the values obtained with Koopmans' [19] theorem and also to the results obtained from some semiempirical calculations. The spacing and ordering of the levels are very similar to the ones obtained in a PPP calculation [1]. The CNDO method gives a different ordering of the highest occupied levels but still gives quite a good overall agreement in the level pattern [3]. The extended Hückel calculations [2, 4] on the other hand give a less satisfactory agreement with the *ab initio* calculation.

The results support the suggestion made by Gouterman [20], that the two highest occupied MOs are well separated from the remainder. The computed ionization potential is 8.00 eV according to the ΔSCF calculation, and 8.46 eV if

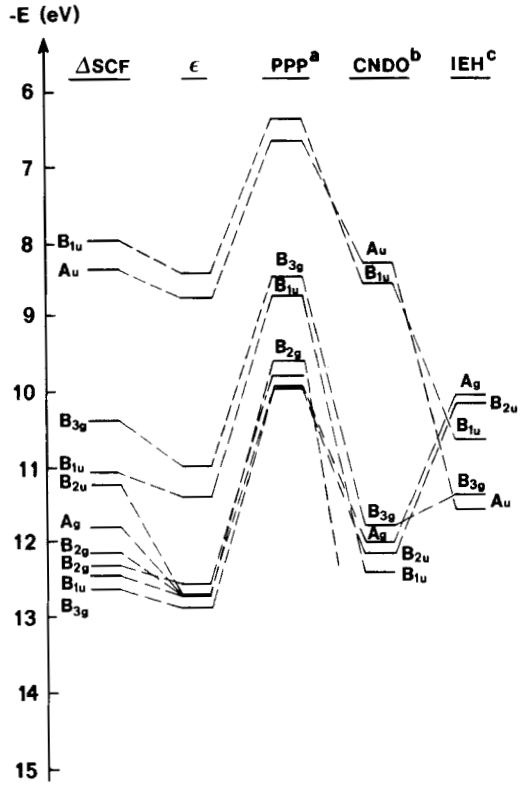

Figure 3. Energy levels for the outer orbitals in porphin. A comparison between Koopmans' theorem, ΔSCF, and some semiempirical calculations. [a] Ref. [1]. [b] Ref. [3]. [c] Ref. [4].

TABLE IV. Core orbital energies for the geometry number 4. The labelling of atoms is the same as in Figure 1. The 'primed' symbols C1′C2′ and N′ refer to the pyrrol fragment with a central hydrogen atom covalently attached to it.

centre	-ε (eV)
N'	427.50
N	424.38
C1'	312.47
C1	311.48
C2'	311.12
C5	311.11
C2	310.32

TABLE V. Eigenvalues and symmetries for the lowest empty molecular orbitals.

Symmetry	ε (eV)
B_{2g}	-1.84
B_{3g}	-1.41
A_u	1.36
B_{1u}	3.91
B_{2g}	5.05
B_{3g}	5.41
A_u	5.72
A_u	7.67
B_{2g}	7.79

Koopmans' theorem is used. From experimentally measured anodic half-wave potentials an I.P. of 7.42 eV has been predicted [21].

The core orbital energies for the geometry number 4 are given in Table IV. The energy splitting for the symmetry-related core levels is quite insignificant (<0.01 eV) and therefore only one value is given for each centre. Table V presents eigenvalues and symmetries for the first few virtual orbitals.

Summary

The positions of the central protons in the free porphin base have been optimized. The lowest energy was found for a structure with D_{2h} symmetry and an N—H distance of 1.025 Å. The NH bending frequencies obtained are similar to those found in hydrogen-bonded systems.

The computed orbital energies show a pattern similar to those obtained from PPP and CNDO callculations. ΔSCF calculations show quite different reorganization energies for the different ionized states, but retain the overall pattern given by the one electron levels.

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